

Cloud Computing Final Project

GROUP # 15

ALI ABDULLAH

ASRA ANEES

FAIZAN SHABEER

Introduction:

Welcome to the development of a highly scalable and dynamic e-commerce website named “Refords” using the popular WordPress platform and WooCommerce plugin. Our goal was to provide a robust solution for a vendor seeking to expand their online presence and handle increased traffic with ease. Utilizing Amazon Web Services (AWS) EC2 using Ubuntu, we have created a platform that can easily scale with the growth of the vendor's business, including the addition of an **Elastic Load Balancer** for traffic management and MySQL database to store the products record. We are confident that this website will provide a seamless shopping experience for customers and a powerful tool for the vendor to drive their business forward.

Deployment Details:

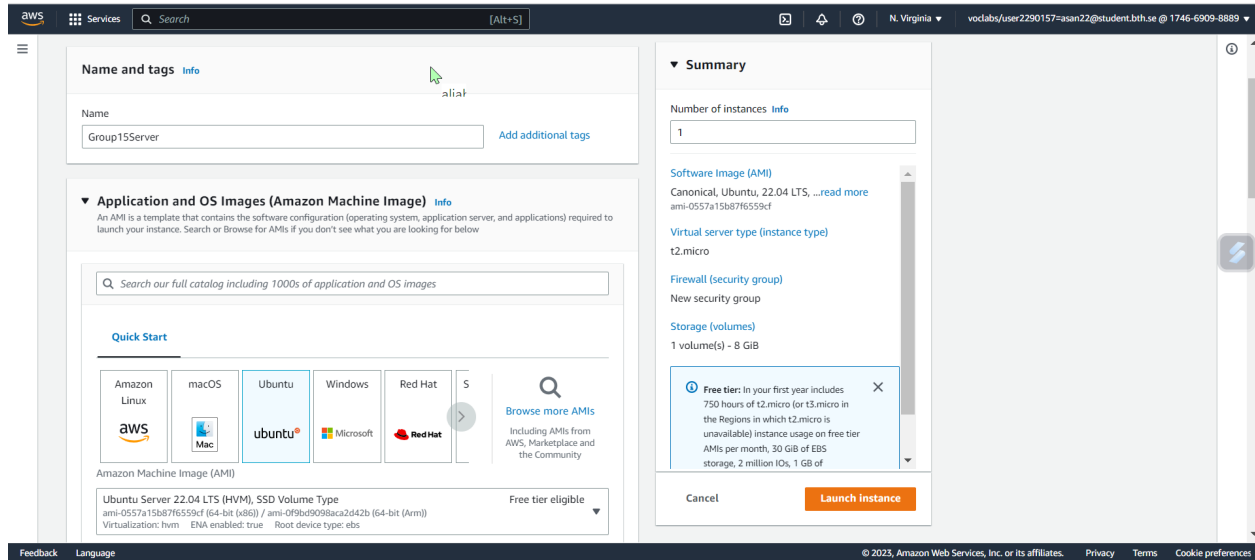
The deployment of the e-commerce website was carried out using AWS **EC2**. The following steps were taken to ensure a successful deployment:

1. Setting up an AWS account and creating an EC2 instance for the website.
2. Setup WordPress and deploy to ubuntu web server.
3. Configuring the Load Balancer to distribute traffic to multiple instances for better performance and availability.
4. Setting up WordPress site's database connection settings for configuration.
5. Testing & Monitoring to verify functionality and performance measurement.

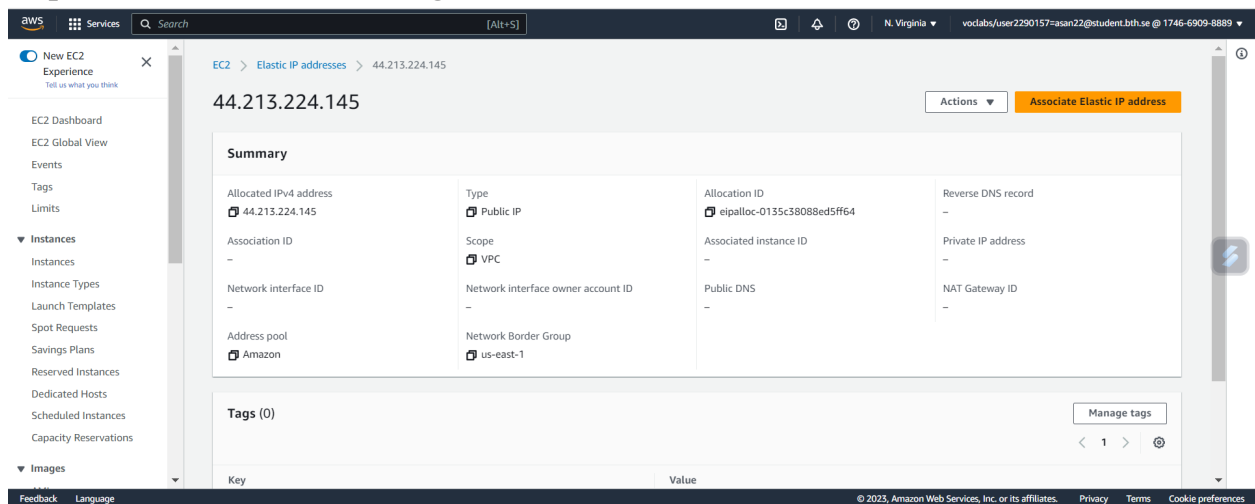
All necessary software and security updates were also managed through the AWS EC2 to ensure the website remains up-to-date and secure. The deployment process was thoroughly tested to guarantee a seamless experience for both the vendor and customers. With the combination of AWS EC2 and the powerful WordPress and WooCommerce platform, we are confident in the scalability and reliability of the e-commerce website.

The following steps are given below:

Step 1: Select Ubuntu



Step 2: Elastic IP created using EC2 instance



Step 3: Elastic IP associated

Elastic IP address: 44.213.224.145

Resource type
Choose the type of resource with which to associate the Elastic IP address.

☒ Instance
☐ Network interface

⚠ If you associate an Elastic IP address with an instance that already has an Elastic IP address associated, the previously associated Elastic IP address will be disassociated, but the address will still be allocated to your account. [Learn more](#)

If no private IP address is specified, the Elastic IP address will be associated with the primary private IP address.

Instance

Private IP address
The private IP address with which to associate the Elastic IP address.

Reassociation
Specify whether the Elastic IP address can be reassociated with a different resource if it already associated with a resource.
☐ Allow this Elastic IP address to be reassociated

Associate

Step 4: Elastic IP address has been associated successfully with instances.

Elastic IP address associated successfully.
Elastic IP address 44.213.224.145 has been associated with instance i-Oebb3898deae5564b

Elastic IP addresses (1/1)

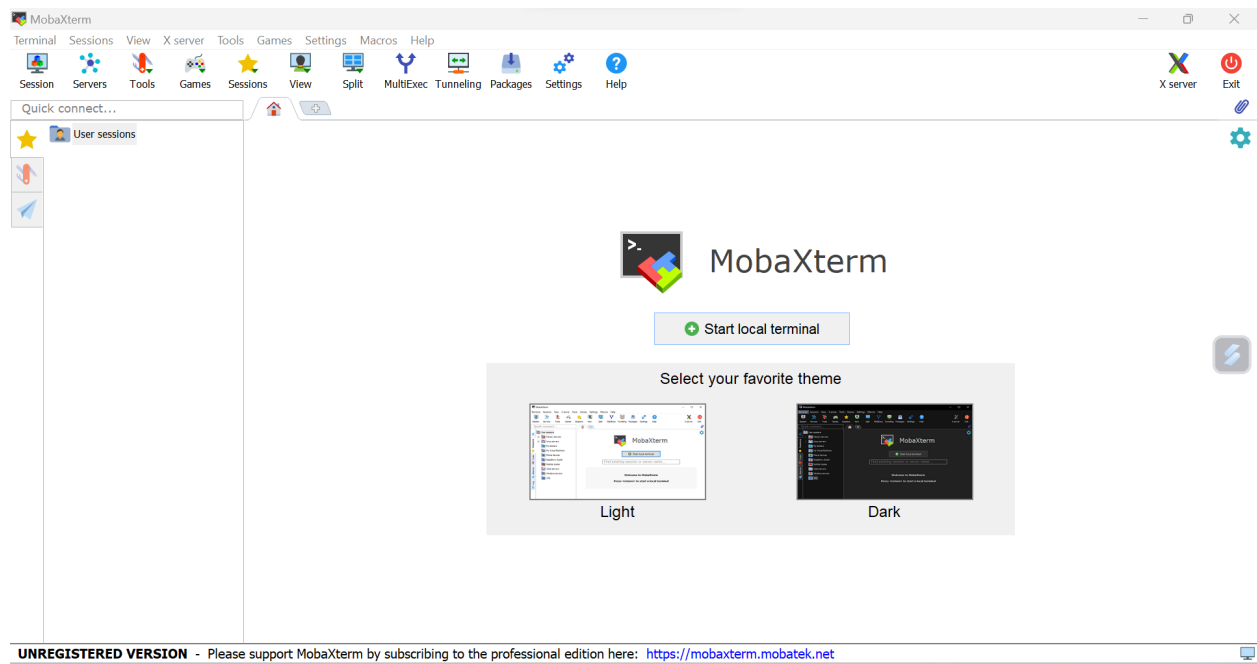
	Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record	Associated instance ID
☒	-	44.213.224.145	Public IP	eipalloc-0135c38088ed5ff64	-	i-Oebb3898deae5564b

44.213.224.145

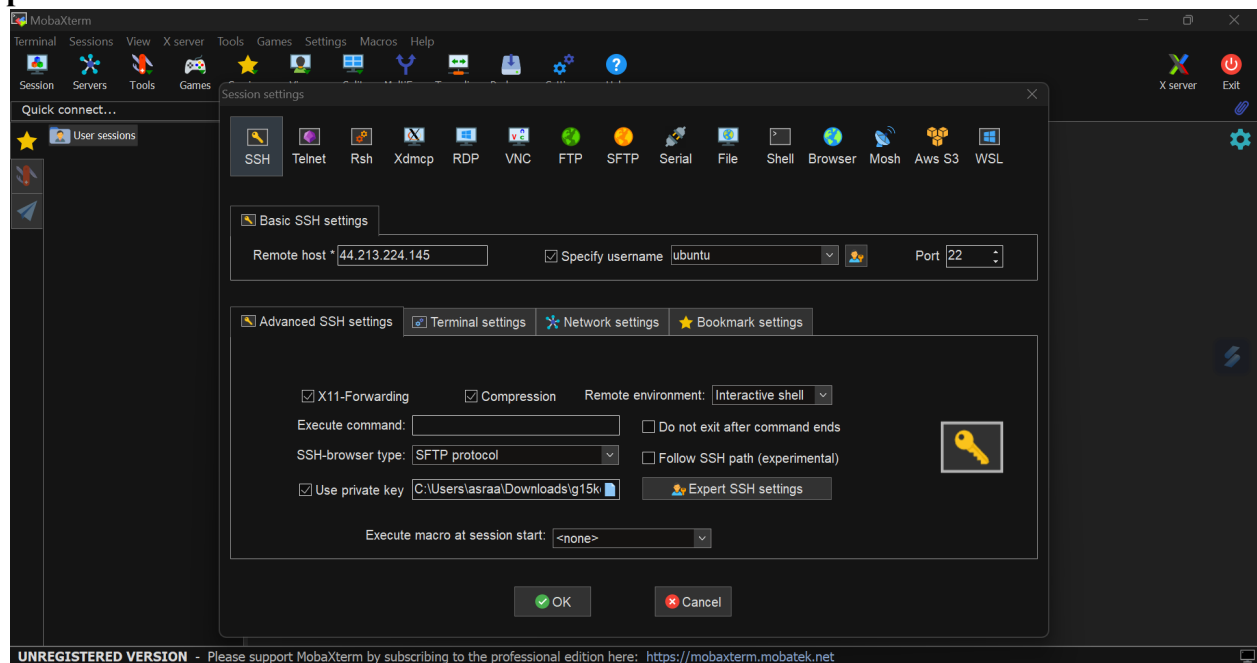
Summary

Summary

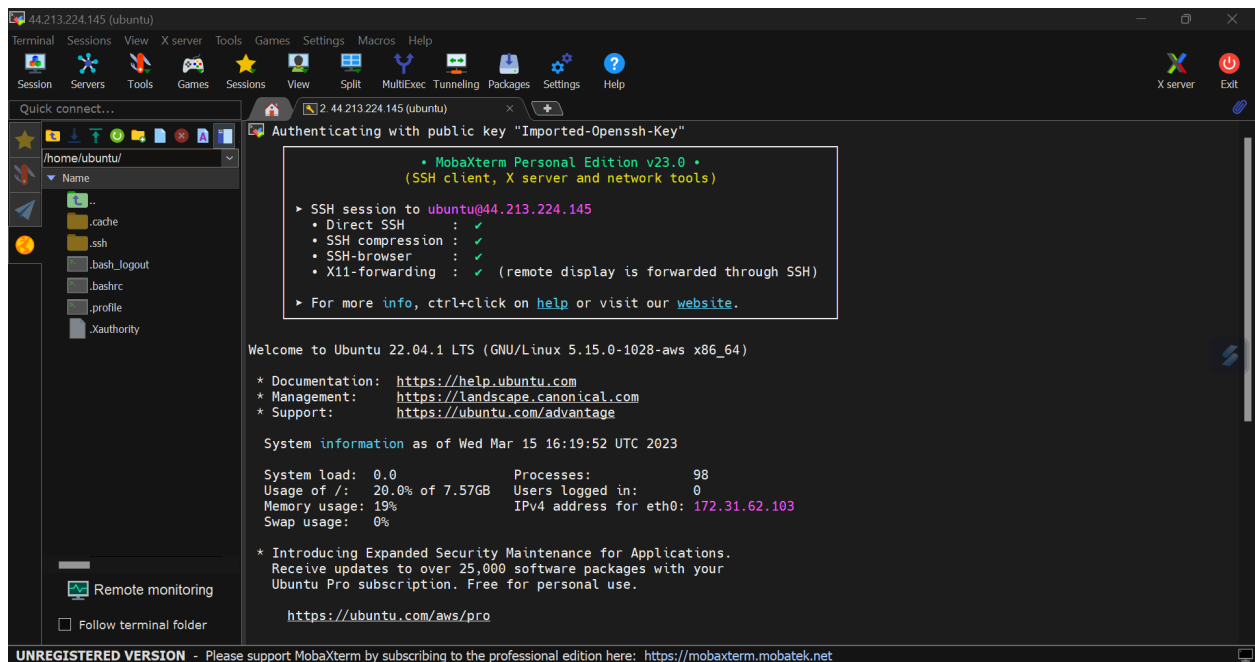
Step 5: Download MobaXterm



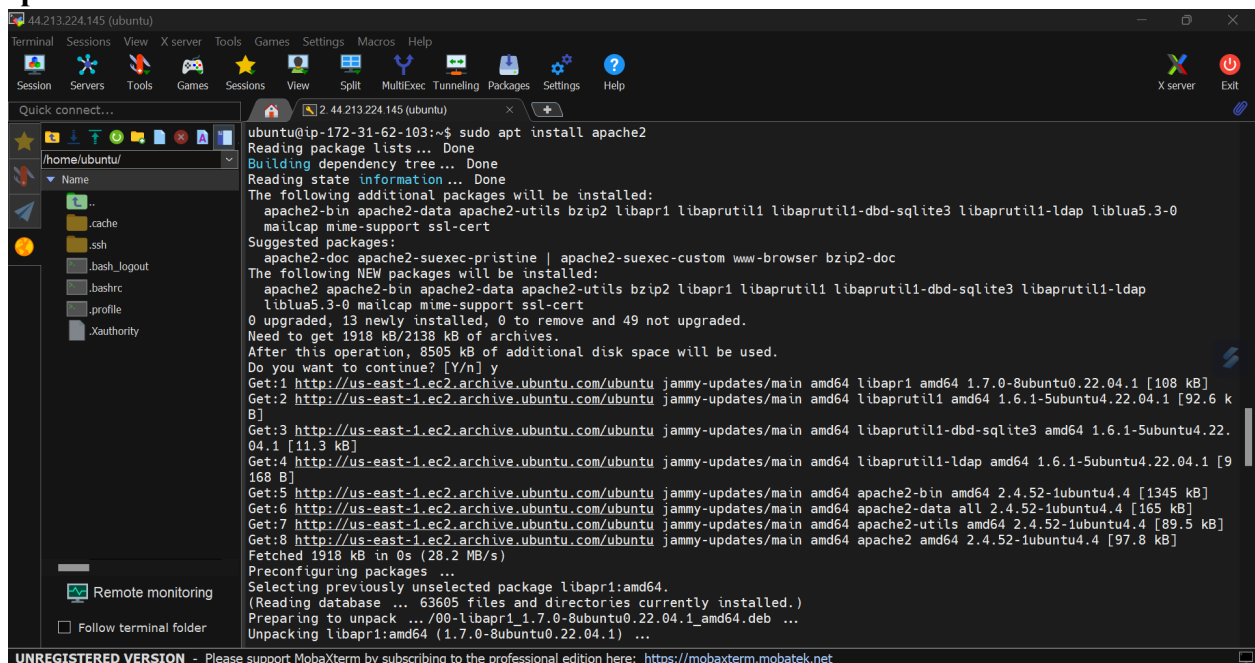
Step 6: Connecting ssh client by providing elastic ip, username, and credentials key pair.



Step 7: Successfully connected to a virtual computer.



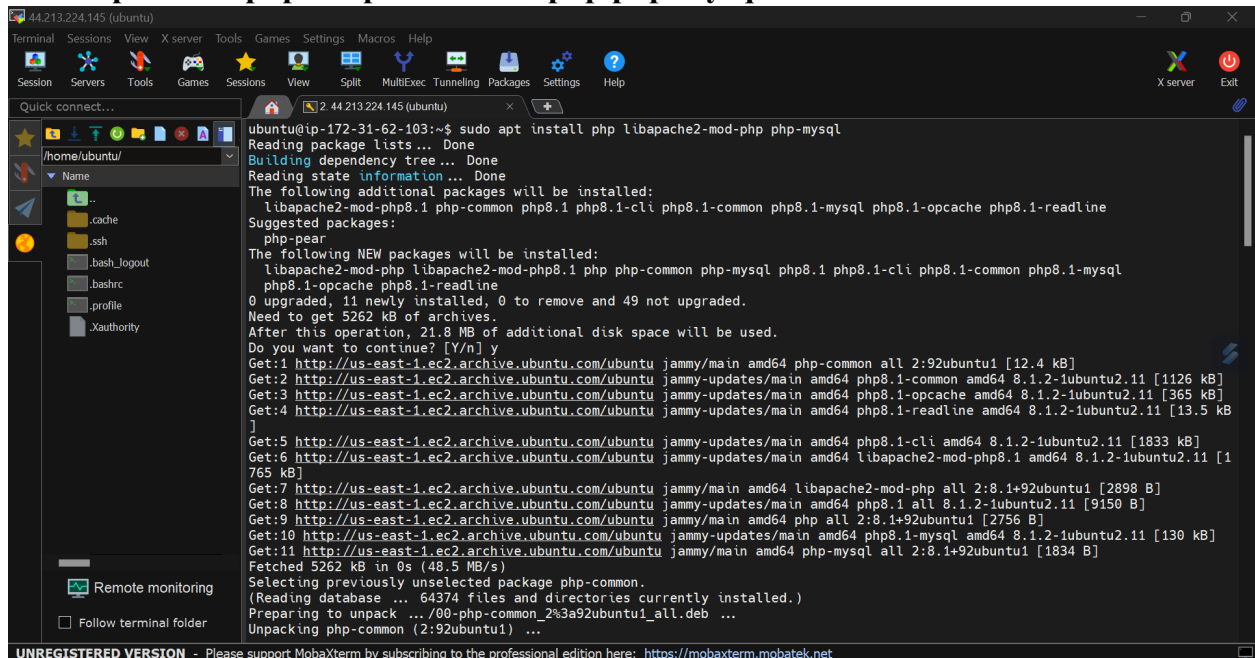
Step 8: Installing apache web server on our instance using command “sudo apt install apache2”.



Step 9: Successful install apache on our instance.



Step 10: Installing php run time and mysql connector on our server. The command is “sudo apt install php libapache2-mod-php php-mysql”.



Step 11: Installing MySQL server for our database

```

ubuntu@ip-172-31-62-103:~$ sudo apt install mysql-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libbcbi-fast-perl libbcbi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7 libfcgi-bin libfcgi-perl
  libfcgi0ldbl libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl
  libio-html-perl liblwp-mediatypes-perl libmecab2 libprotobuf-lite23 libtimedate-perl liburi-perl mecab-ipadic
  mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0 mysql-common mysql-server-8.0 mysql-server-core-8.0
Suggested packages:
  libdata-dump-perl libipc-sharedcache-perl libbusiness-isbn-perl libwww-perl mailx tinycsa
The following NEW packages will be installed:
  libbcbi-fast-perl libbcbi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7 libfcgi-bin libfcgi-perl
  libfcgi0ldbl libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl
  libio-html-perl liblwp-mediatypes-perl libmecab2 libprotobuf-lite23 libtimedate-perl liburi-perl mecab-ipadic
  mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0 mysql-common mysql-server mysql-server-8.0
  mysql-server-core-8.0
0 upgraded, 28 newly installed, 0 to remove and 49 not upgraded.
Need to get 29.5 MB of archives.
After this operation, 242 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 mysql-common all 5.8+1.0.8 [7212 B]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 mysql-client-core-8.0 amd64 8.0.32-0ubuntu0.22.04.2 [2677 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 mysql-client-8.0 amd64 8.0.32-0ubuntu0.22.04.2 [22.7 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 libevent-pthreads-2.1-7 amd64 2.1.12-stable-1build3 [7642 B]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 libmecab2 amd64 0.996-14build9 [199 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libprotobuf-lite23 amd64 3.12.4-1ubuntu7.22.04.1 [209 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 mysql-server-core-8.0 amd64 8.0.32-0ubuntu0.22.04.2 [17.5 MB]

```

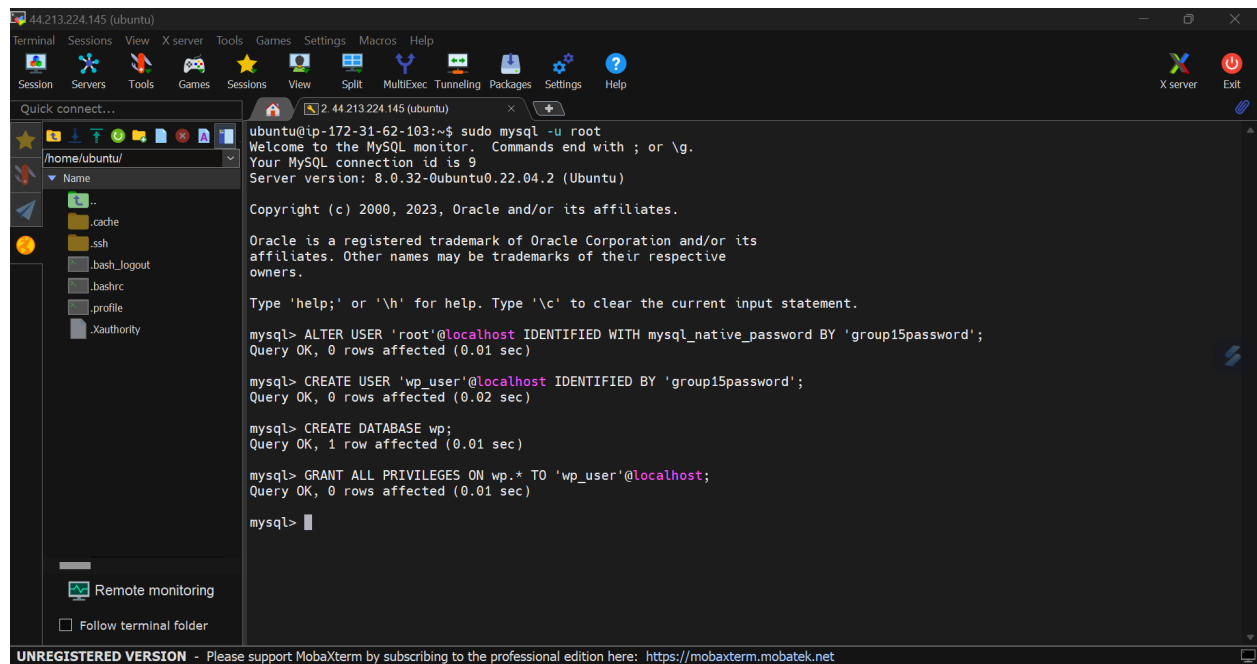
Step 12: Configuration on mysql server: changing mysql authentication plugin to mysql native password; so that we can login to the mysql server with the normal password. It is required in order to install wordpress.

To do that first login to mysql prompt with root user(using "sudo mysql -u root"), then changing password using ("ALTER USER 'root'@localhost IDENTIFIED WITH mysql_native_password BY 'group15password' ").

Then create another sub-user other than root for wordpress by using ("CREATE USER 'wp_user'@localhost IDENTIFIED BY 'group15password';").

And we also need a separate database to use with wordpress, we will create separate database by using (" CREATE DATABASE wp;")

Finally we will assign all the privileges to this user by doing (" GRANT ALL PRIVILEGES ON wp.* TO 'wp_user'@localhost;")



A screenshot of the MobaXterm application window. The terminal pane shows a user logging into an Ubuntu server at IP 172-31-62-103 and running a series of MySQL commands. The left sidebar shows a file explorer for the /home/ubuntu directory. The top menu bar includes Terminal, Sessions, View, X server, Tools, Games, Settings, Macros, and Help. The bottom status bar indicates it is an 'UNREGISTERED VERSION'.

```
ubuntu@ip-172-31-62-103:~$ sudo mysql -u root
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.0.32-0ubuntu0.22.04.2 (Ubuntu)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> ALTER USER 'root'@localhost IDENTIFIED WITH mysql_native_password BY 'group15password';
Query OK, 0 rows affected (0.01 sec)

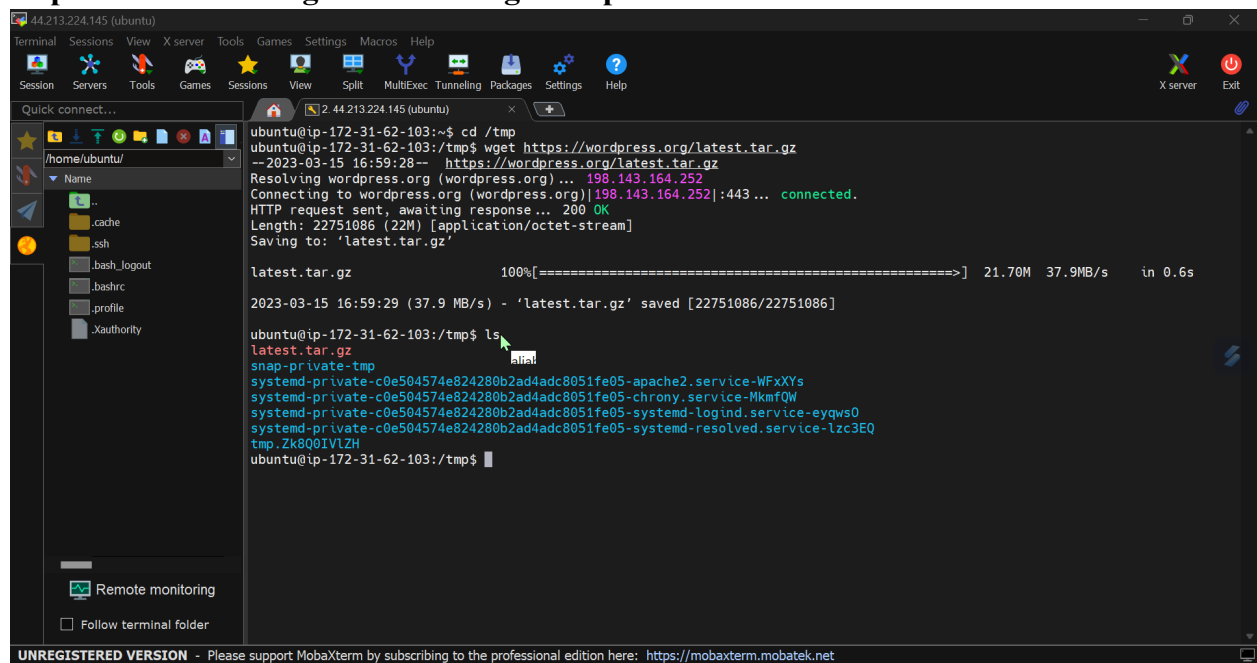
mysql> CREATE USER 'wp_user'@localhost IDENTIFIED BY 'group15password';
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE DATABASE wp;
Query OK, 1 row affected (0.01 sec)

mysql> GRANT ALL PRIVILEGES ON wp.* TO 'wp_user'@localhost;
Query OK, 0 rows affected (0.01 sec)

mysql>
```

Step 13: Downloading and installing wordpress.



A screenshot of the MobaXterm application window. The terminal pane shows the user downloading the latest WordPress tar.gz file from the official website using the 'wget' command. The download progress is shown as 100% complete. The left sidebar shows the same file explorer as the previous screenshot. The bottom status bar indicates it is an 'UNREGISTERED VERSION'.

```
ubuntu@ip-172-31-62-103:~$ cd /tmp
ubuntu@ip-172-31-62-103:/tmp$ wget https://wordpress.org/latest.tar.gz
--2023-03-15 16:59:28-- https://wordpress.org/latest.tar.gz
Resolving wordpress.org (wordpress.org) ... 198.143.164.252
Connecting to wordpress.org (wordpress.org)|198.143.164.252|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 22751086 (22M) [application/octet-stream]
Saving to: 'latest.tar.gz'

latest.tar.gz      100%[=====] 21.70M  37.9MB/s   in 0.6s

2023-03-15 16:59:29 (37.9 MB/s) - 'latest.tar.gz' saved [22751086/22751086]

ubuntu@ip-172-31-62-103:/tmp$ ls
latest.tar.gz
snap-private-tmp
systemd-private-c0e504574e824280b2ad4adc8051fe05-apache2.service-WFxXYs
systemd-private-c0e504574e824280b2ad4adc8051fe05-chrony.service-MkmfQW
systemd-private-c0e504574e824280b2ad4adc8051fe05-systemd-logind.service-eyqws0
systemd-private-c0e504574e824280b2ad4adc8051fe05-systemd-resolved.service-lzc3EQ
tmp.Zk8Q0IvLZH
ubuntu@ip-172-31-62-103:/tmp$
```

Step 14: Moved extracted wp files to ip followed by wordpress.

```
wordpress/wp-admin/js/common.js
wordpress/wp-admin/js/set-post-thumbnail.min.js
wordpress/wp-admin/js/postbox.min.js
wordpress/wp-admin/js/color-picker.js
wordpress/wp-admin/js/password-strength-meter.js
wordpress/wp-admin/js/customize-nav-menus.js
wordpress/wp-admin/js/editor-expand.js
wordpress/wp-admin/js/code-editor.min.js
wordpress/wp-admin/js/set-post-thumbnail.js
wordpress/wp-admin/options-permalink.php
wordpress/wp-admin/widgets.php
wordpress/wp-admin/setup-config.php
wordpress/wp-admin/install.php
wordpress/wp-admin/admin-header.php
wordpress/wp-admin/post-new.php
wordpress/wp-admin/themes.php
wordpress/wp-admin/options-reading.php
wordpress/wp-trackback.php
wordpress/wp-comments-post.php
ubuntu@ip-172-31-62-103:/tmp$ ls
latest.tar.gz
snap-private-tmp
systemd-private-c0e504574e824280b2ad4adc8051fe05-apache2.service-WFXYs
systemd-private-c0e504574e824280b2ad4adc8051fe05-chrond.service-MkmfQW
systemd-private-c0e504574e824280b2ad4adc8051fe05-systemd-logind.service-eyqws0
systemd-private-c0e504574e824280b2ad4adc8051fe05-systemd-resolved.service-lzc3EQ
tmp.Zk8Q0IVLZH
wordpress
ubuntu@ip-172-31-62-103:/tmp$ sudo mv wordpress/ /var/www/html/
ubuntu@ip-172-31-62-103:/tmp$ cd /var/www/html/
ubuntu@ip-172-31-62-103:/var/www/html$ ls
index.html  wordpress
ubuntu@ip-172-31-62-103:/var/www/html$
```

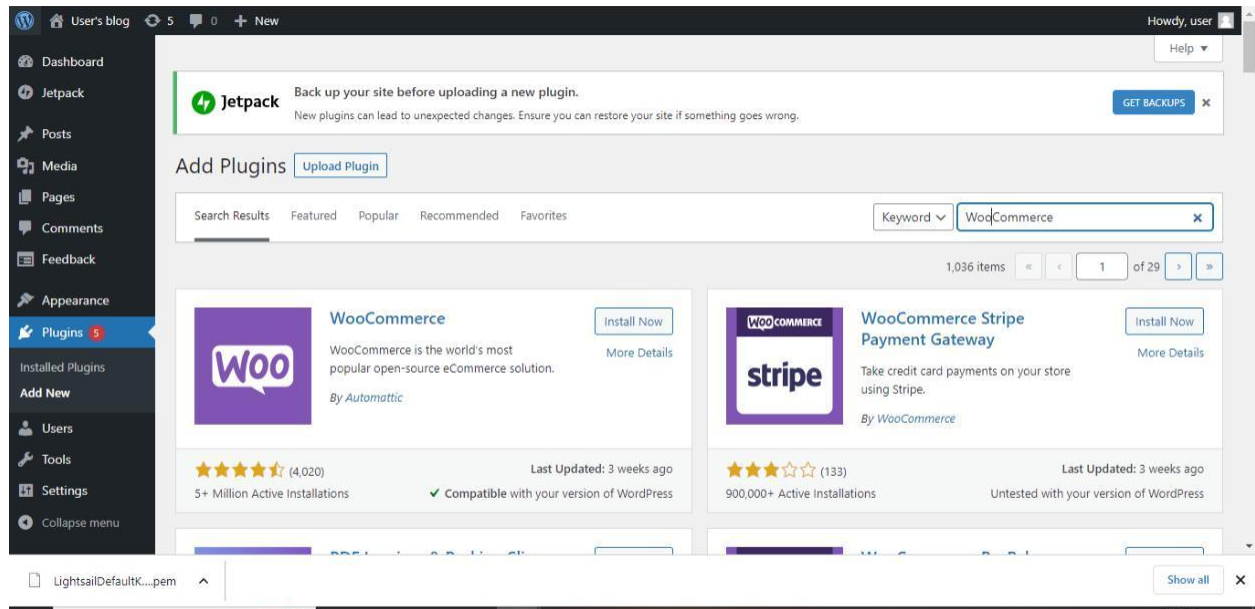
UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: <https://mobaxterm.mobatek.net>

Step 15: Configuring wordpress

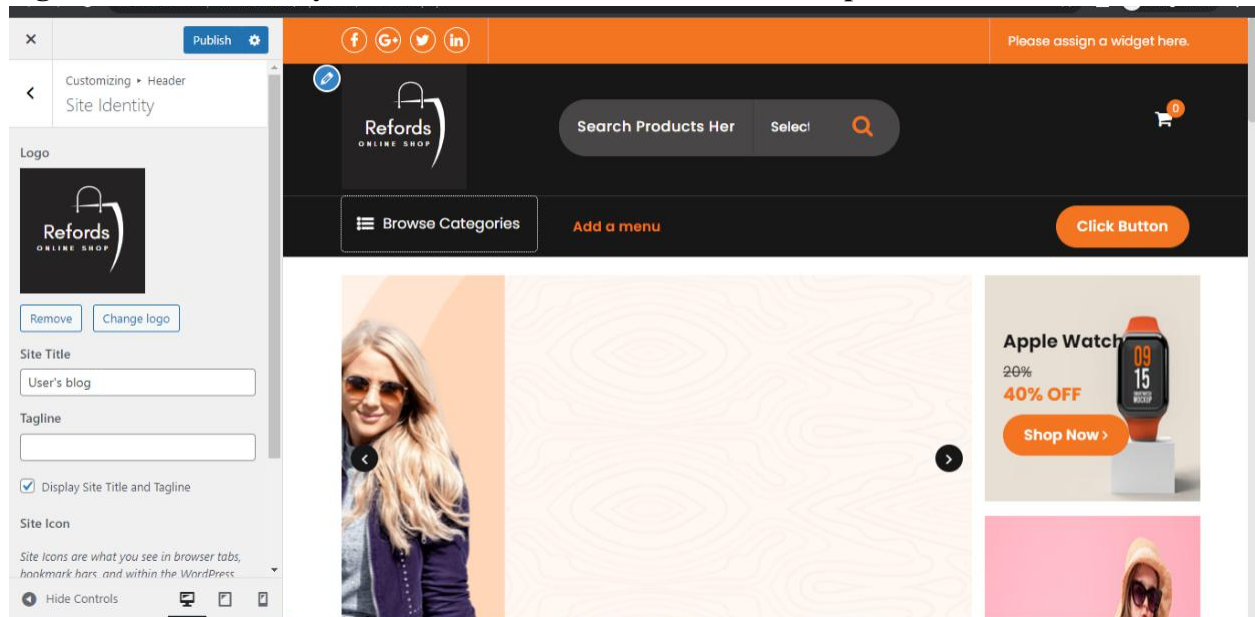
Below you should enter your database connection details. If you are not sure about these, contact your host.

Database Name	wp	The name of the database you want to use with WordPress.
Username	wp_user	Your database username.
Password	group15password	Your database password.
Database Host	localhost	You should be able to get this info from your web host, if localhost does not work.
Table Prefix	wp_	If you want to run multiple WordPress installations in a single database, change this.

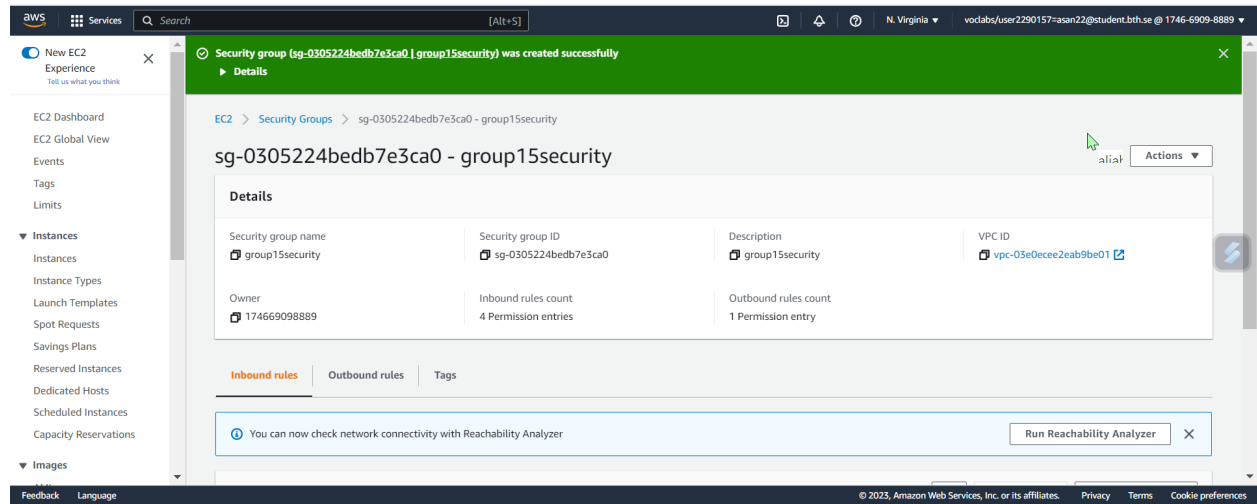
Step 16: Installing WooCommerce



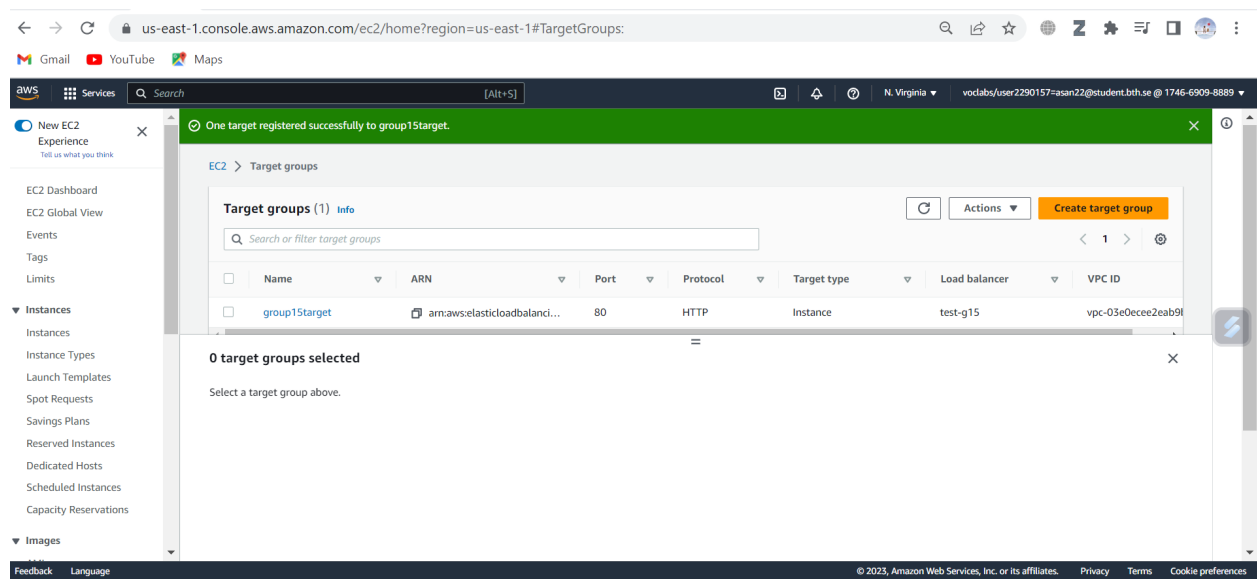
Step 17: Successfully installed WooCommerce plugin and created a custom logo for the website by its name and added the relative template.



Step 18: Creating Security Group for Load balancer creation



Step 19: Created Target Group and associated it with the Load Balancer.



Step 20: Using “sudo apt-get install stress” and then applying “stress -c 10” we stressed the cpu and then monitored it’s utilization.

```
44.213.224.145 (ubuntu)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quick connect...
/home/ubuntu/
Name
.cache
.local
.ssh
.bash_history
.bash_logout
.bashrc
.profile
.sudo_as_admin_successful
.xauthority
Remote monitoring
Follow terminal folder

E: Invalid operation get
ubuntu@ip-172-31-62-103:~$ sudo apt-get install stress
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
stress
0 upgraded, 1 newly installed, 0 to remove and 49 not upgraded.
Need to get 18.4 kB of archives.
After this operation, 52.2 kB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/universe amd64 stress amd64 1.0.5-1 [18.4 kB]
Fetched 18.4 kB in 0s (600 kB/s)
Selecting previously unselected package stress.
(Reading database ... 65125 files and directories currently installed.)
Preparing to unpack .../stress_1.0.5-1_amd64.deb ...
Unpacking stress (1.0.5-1) ...
Setting up stress (1.0.5-1) ...
Processing triggers for man-db (2.10.2-1) ...
Scanning processes ...
Scanning linux images ...

Running kernel seems to be up-to-date.

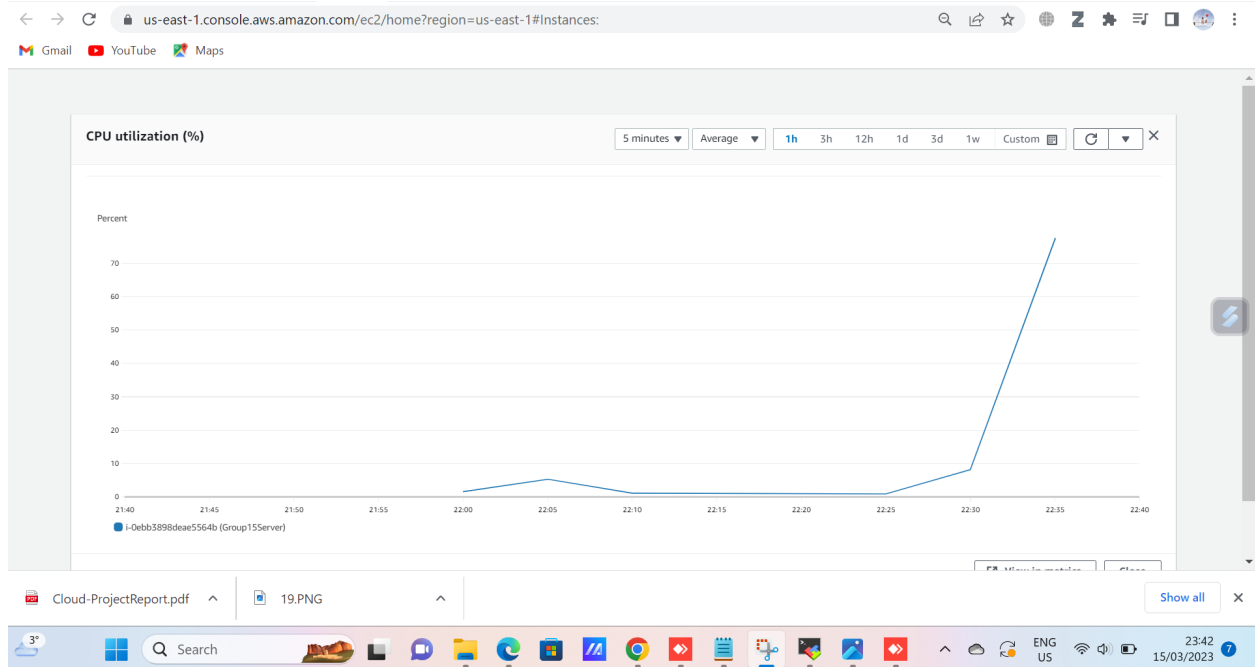
No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-62-103:~$ stress -c 10
stress: info: [2158] dispatching hogs: 10 cpu, 0 io, 0 vm, 0 hdd
```

Step21: This is the Cpu Utilization after applying stress to our instance.



Motivation: We Selected WooCommerce as an ecommerce platform because Wordpress provides the most popular plugins to integrate and we selected Ubuntu apache as our web server because Ubuntu is based on the Linux kernel and Linux provides the most powerful shell.

Conclusion:

In conclusion, the development and deployment of the e-commerce website using WordPress and WooCommerce on AWS EC2 was a success. The website provides a user-friendly interface for customers, a powerful tool for the vendor to manage their business, and the ability to easily scale and adapt to growth. With the Load Balancer, the vendor can confidently handle increased traffic and expanding product offerings. The website was designed and deployed with security, performance, and reliability in mind, ensuring a positive experience for both the vendor and customers. We are proud to have been a part of this project and look forward to seeing the vendor's continued success in the future.